Vacuum/volume 33/number 1/2/pages 35 to 41/1983
Printed in Great Britain

0042-207X/83/010035-07$03.00/0
Pergamon Press Ltd

P9

Structure and electrical properties of amorphous Ni–P thin films

E Besenyei, B Zsoldos, *Industrial Research Institute for Electronics, HIKI, H-1393 Budapest 62, PO Box 348, Hungary*

and

O Geszti, *Research Institute for Technical Physics of the Hungarian Academy of Sciences, Budapest, Hungary*

Amorphous Ni–P films were prepared by electroless deposition. Electrical properties (sheet resistivity and temperature coefficient) of the resistance layer deposited on ceramic and glass substrates under various conditions were determined. The structure and morphology of the films were investigated by transmission electron microscopy and electron diffraction. A relationship was found between the structure and the electrical properties. In addition the effect of annealing, carried out at different temperatures, on the electrical properties of the resistance layer was studied.

Introduction

Preparation of thin film resistors by electroless deposition has been applied for about 10 yr. This method is favourable compared to vacuum deposition, because no expensive vacuum equipment is needed. The metal deposits from a solution on to the surface of the activated substrate[1]. As the application of this method for preparing thin film resistors is relatively new, there is little data concerning the properties and possible applications of these films. These resistor films are made mainly of nickel alloys with high phosphorous content, because they have an amorphous structure, and in consequence of this they have relatively high specific resistivity and a low temperature coefficient of resistivity.

Although nickel–phosphorous alloys have been widely used recently, there is no generally accepted theory explaining their physical properties. There are contradictions in descriptions of their crystalline structure and in explanations of structural changes during annealing[2–6]. Depending on the preparation techniques, the homogeneity of the nickel–phosphorous alloys may vary, and the changes during annealing of these layers are different for the inhomogeneous films[7].

The aim of our work was to investigate the properties of resistors made by electroless deposition, as a function of different preparation parameters, and to establish the character of changes occurring during annealing in air.

Sample preparation

The resistor films were prepared on glass and ceramic substrates. The glass substrate (Corning 7059) was cleaned in freon degreaser before the chemical treatment.

The ceramic substrate containing 96% alumina, used in thick film technology, was cleaned in alkaline solution before deposition. The activation of the substrates was performed by the classical $SnCl_2$–$PdCl_2$ method[1]. The nickel resistor layer was deposited from the following bath:

$NiCl_2 . 6H_2O$	45 g l^{-1}
NH_4Cl	50 g l^{-1}
$Na_3C_6H_5O_7 . 1/2H_2O$	10 g l^{-1}
$NaH_2PO_2 . H_2O$	11 g l^{-1}.

The bath temperature was 65° and 70°C for ceramic and glass substrates respectively. The deposition time was 45 s for glass substrates and 2 min for ceramic substrates, unless the deposition time was the parameter to be investigated.

In order to investigate the electrical properties of the films, a test resistor pattern was made on plates. The electrical contacts were made by vacuum evaporation through masks, as used in thin film technology. The contact layer consisted of 40–60 nm of Ti, 60–80 nm of NiCr and 250 nm of Au. On substrates containing resistor films and contacts, the test patterns were made by photolithography. The investigation of the effect of altering technological parameters was carried out on similar plates, but the contacts were made of Ni prepared by electroless deposition.

The parameters of the bath were investigated by changing one of the bath parameters and by keeping all other process parameters constant. The reduction of the films was carried out in fresh, unused baths in each case. The temperature of the bath was controlled by a thermostat within ± 0.1°C. The pH of the bath was measured by OP-08083 type glass electrode and OP 208 type gauge. The deposition rate was determined by calculation from the deposited mass and the surface area of the films prepared in 1 hr.

The phosphorous content of the films was determined by two different methods. The phosphorous content of the surface was obtained with an electron microprobe attachment on a scanning electron microscope (type JEOL-JSM-35). The phosphorous concentration in about 1 μm^3 generated volume was determined

by a computer which took into account difference corrections as well. The average phosphorous content of the films was determined by colorimetry with a Labor MIM continuous automatic analyser, type Contiflo.

When the phosphorous content of the film was investigated, the deposition time took 1 h. The film was dissolved in 1:1 nitric acid solution to analyse the phosphorous content.

Experimental methods

The resistivity measurements were carried out with a 'SOLATRON 7065' microprocessor voltmeter. The sheet resistance of the film was calculated with the $R_{\square} = R(1/d)$ formula.

The thermal coefficient of resistance of the resistors was measured by means of temperature controlled equipment. The effect of the technological parameters of the preparation was investigated; determining the electrical parameters after annealing 5 h at 250°C temperature.

The structure of the film was investigated by a JEOL 100 U transmission electron microscope at 100 kV accelerating voltage. For this purpose the Ni-layer was deposited on the surface of an amorphous SiO_x film supported by a platinum grid. The crystalline structure was investigated by electron diffraction.

In order to investigate the changes during annealing, samples were annealed in air at temperatures between 120–300°C, and the corresponding resistivity change was recorded continuously. In all tests, the samples were put into a furnace previously heated to the annealing temperature.

Experimental results and discussion

The influence of preparation technology was mainly investigated for ceramic substrates. The resistor properties were strongly influenced by temperature, pH and the time of deposition.

The pH of the bath was changed by feeding hydrochloric acid or ammonium hydroxide into it. Below pH 3, deposition of a continuous film did not occur, between pH 6 and pH 8, after 2 min of deposition, a continuous film was not obtained. If the time of deposition was longer, in this pH-range the nickel layer was powdery with poor adhesion. Below pH 3 the nickel film must be dissolved; at pH 6 the solubility of nickel phosphite lowers and the solution may be decomposed, resulting in a coarse coating. At pH above 8 an intensive deposition begins to occur. Figure 1 shows the electrical properties of resistor films as a function of the pH of the bath. The preparation took 2 min at 65°C in the bath mentioned above. The figure shows that the resistivity of the film decreases with the increase of the pH value. This may be caused by the increase of deposition rate as shown in Figure 2.

It is surprising that the temperature coefficient of the resistors increases to a great extent when the pH of the bath is increasing. At pH 9 the TCR value of the film is in the range of the TCR of the pure metal.

Figure 3 shows the changes of the electrical properties of the resistor films as a function of the time of deposition at pH 4.3, and Figure 4 shows the same dependence at pH 8.2. The figures show

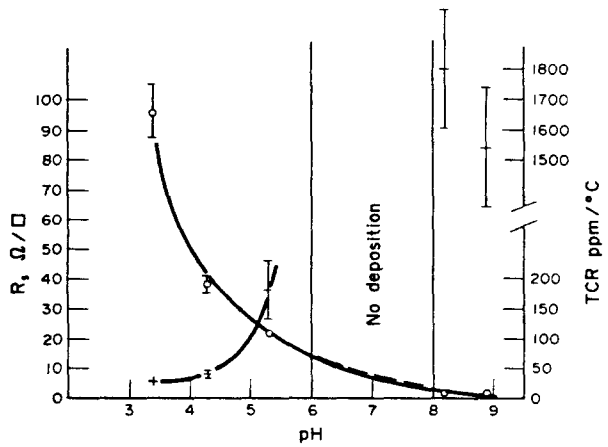


Figure 1. Electrical properties of the resistor films as a function of the pH of the bath.

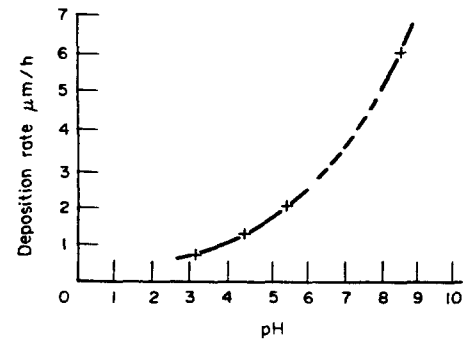


Figure 2. Deposition rate of the Ni-layer as a function of the pH of the bath.

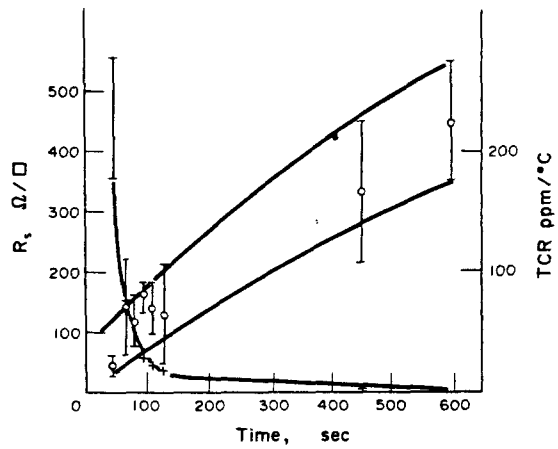


Figure 3. Sheet resistance (+) and TCR value (°) as a function of the deposition time at pH 4.3.

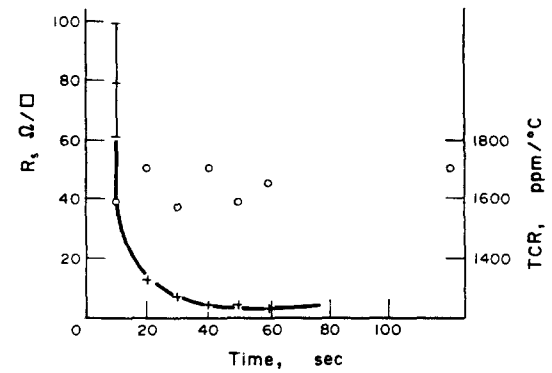
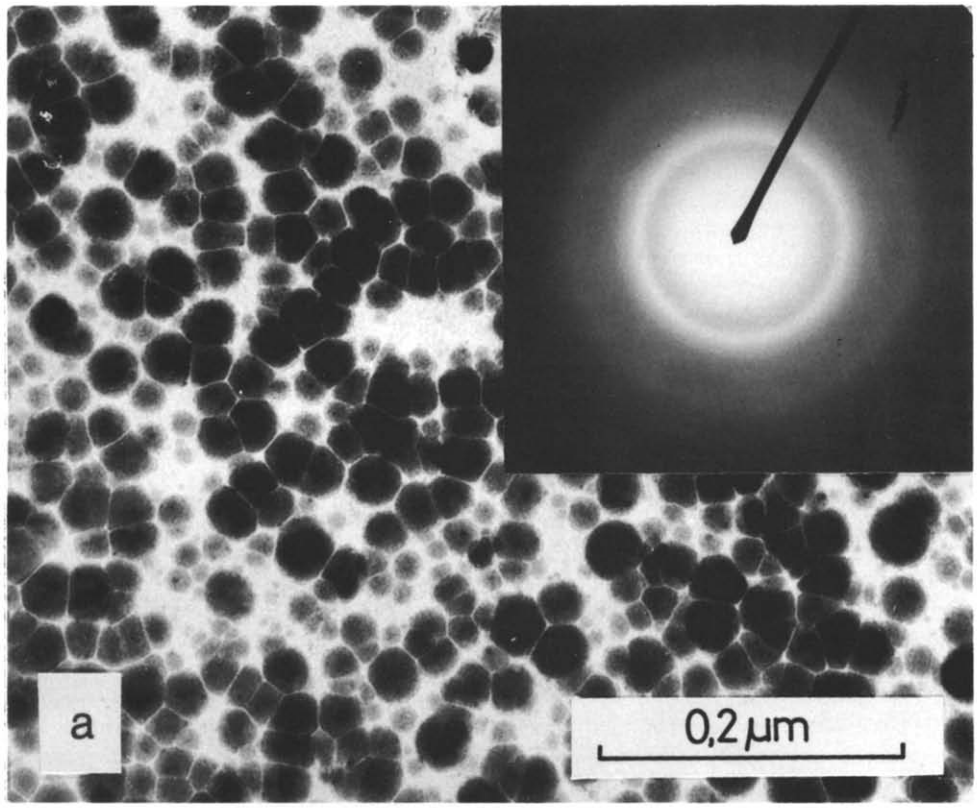


Figure 4. Sheet resistance (+) and TCR value (°) as a function of the deposition time at pH 8.2.



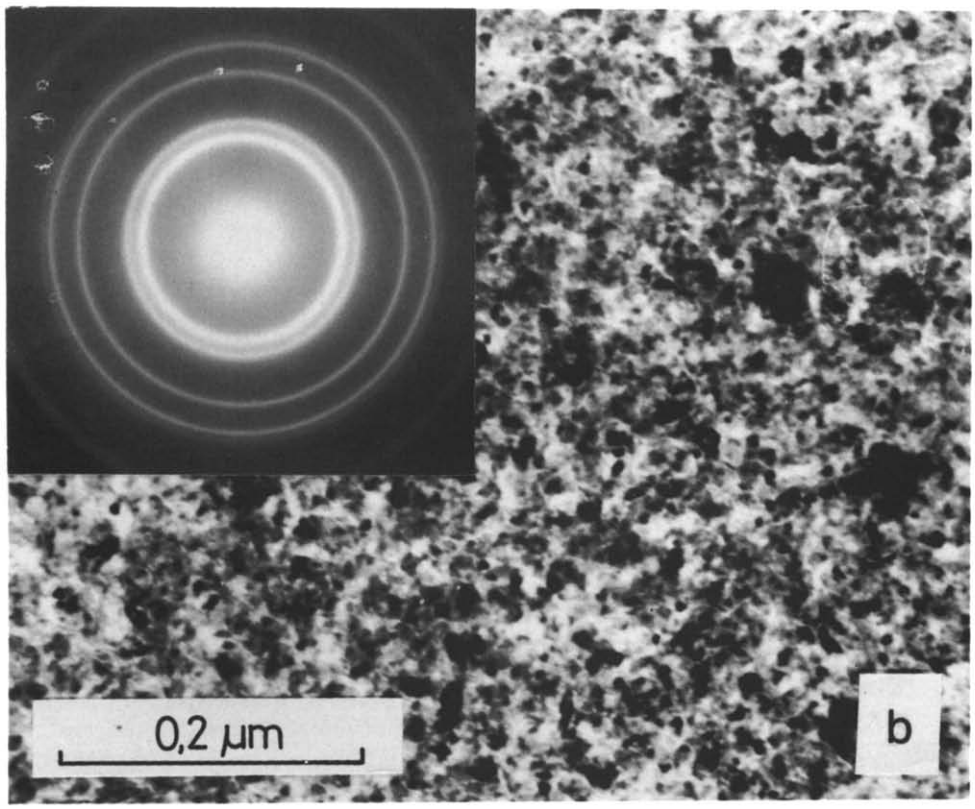


Figure 5. TEM picture and electron diffractions of resistor films deposited in acidic (a) and alkaline (b) baths.

that both by the use of acidic and alkaline baths the resistivity of the prepared resistors decreases with increasing deposition time. It can also be seen that the standard deviation of resistivity values is bigger at higher resistivity values i.e. shorter deposition times. This can be explained by the mechanism of film structure building. During the activation process, activation centres are created. The size of these centres grows during the film build-up and, gradually, continuous films are formed.

A comparison between Figures 3 and 4 shows that films made in an alkaline bath have a TCR value in the range of that for pure metals even if the resistivity is high, but films made in an acidic bath have lower TCR value even if the sheet resistivity is low.

The films prepared from acidic and alkaline baths have different structures, as can be seen by electron microscopy. Films prepared in an acidic bath have an amorphous structure, while the structure of a film made from an alkaline bath has a crystalline character. In order to determine the reason for this structural difference, the phosphorous content of the film was determined as a function of the pH of the bath.

Similar phosphorous contents were found by both methods and results of the experiment are shown in Figure 6. This figure shows that the phosphorous content decreases with an increase in the pH value. The effect of the phosphorous content on the temperature coefficient is very important since the structure of the film and its electrical properties are determined by the phosphorous content. Our results are in accordance with other reported data[8] which suggests that, if the phosphorous content of the film is over 7%, the film structure is amorphous while, below 7% phosphorous content, the film structure is crystalline. This structural difference also explains the difference in TCR values. The crystalline metallic layer has a high positive TCR value while, in an amorphous film, electron motion is more difficult, the increase in temperature has only a slight influence on the resistivity and, consequently, the TCR value is lower.

The effect of the bath temperature was investigated between 55 and 90°C. Below 55°C no continuous film developed. Figure 7 shows the results of the test. The figure shows that the resistivity decreases by increasing the temperature. The rate of autocatalytical reduction (like all chemical reactions) depends considerably on the temperature. Therefore the resistivity decrease results from the increase of the reaction rate. Figure 7 shows well that the TCR value is negative at the beginning of the deposition; then it turns to a positive value and increases with increasing temperature.

The initial negative TCR may be caused by the discontinuous island-structure film. Measurement of the phosphorous content of the films shows that it decreases (from 12.5 to 9.0%) with increasing temperature. This decrease in phosphorous content

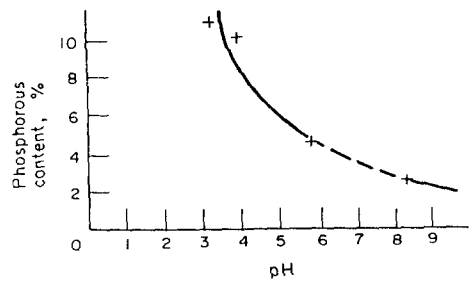


Figure 6. Phosphorous content of the nickel films as a function of the pH of the bath.

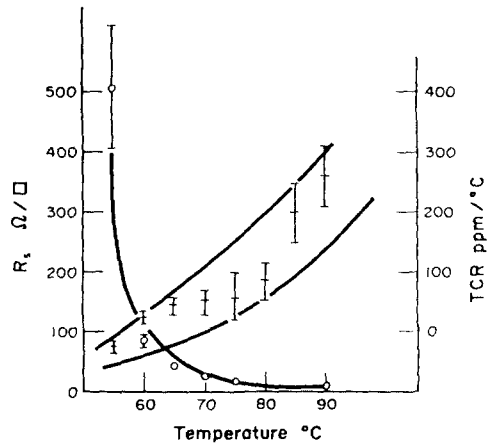


Figure 7. Electrical properties of nickel resistor films as a function of the bath temperature.

may also cause an increase in the TCR value, although the increase of film thickness shifts the TCR value in the positive direction even for constant phosphorous content, as seen in the case of films prepared in an acidic bath for different periods of time.

For characterization of thin film resistors a suitable method is the measurement of changes occurring as a result of the annealing and electrical load. In this test the resistors were annealed at different temperatures (150°C, 200°C, 250°C and 300°C) for different periods of time, and the resistivity changes during annealing were recorded.

Figure 8 shows the resistivity change of nickel films prepared on glass substrates as a function of annealing time. The annealing was carried out at 150°C, 200°C and 300°C. The ordinate is the relative resistivity change ($\Delta R/R$; %), the abscissae is the time of annealing on a logarithmic scale. By comparing curves obtained when annealing at different temperatures, it is seen that the character of the changes occurring during annealing is similar.

In the first 5 to 10 min of the annealing the sample reaches the temperature of the furnace. In this period, the length of which decreases with increasing temperature, the resistivity increases as a result of the positive TCR value of the sample. This short period which is not visible in all the curves because of the logarithmic scale is followed by a period of resistivity decrease. After a minimum is reached, the resistivity increases again. Later on a process decreasing the resistivity becomes dominant again, so after passing through a maximum the resistivity starts to decrease and, finally, it reaches a near constant value.

By annealing at 200°C or below only the first and the second sections of the annealing curve (the initial increase and subsequent decrease in resistivity) can be seen. The third and fourth sections could be examined only during annealing at 250°C and 300°C with a test life of less than 100 h duration.

In order to investigate the mechanism of the processes corresponding to the different sections of the annealing curve, the films were examined by transmission electron microscopy after annealing at 300°C for different periods. The deposition time was 45 s and the films were deposited on amorphous SiO_x evaporated on a platinum grid. The pictures are given on Figure 9. The letters on the pictures indicate the annealing time similarly identified on the annealing curves. With the help of the structural investigations, the processes in the different sections of the annealing curve could be identified.

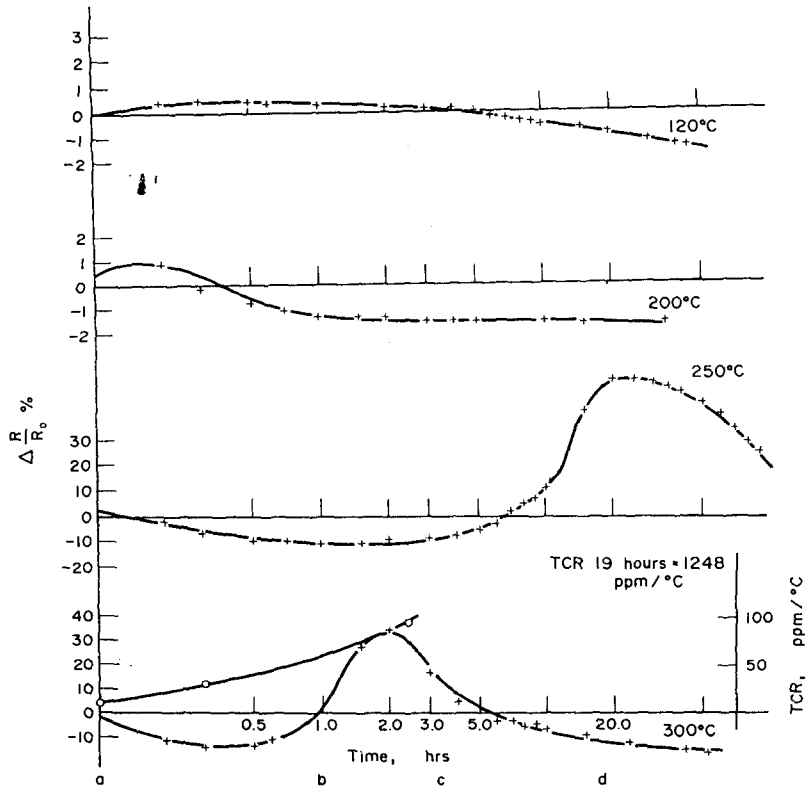


Figure 8. Relative resistivity change of nickel films deposited on glass substrates as a function of the annealing time.

The initial film has an amorphous structure, where the film consists of 20–24 nm dia isles of irregular shape. The density of these islands is not homogeneous. At the beginning a compound phase cannot be identified. In the first and second stage of the annealing (increase in resistivity followed by a decrease) no change in the structure can be seen. The decrease of the resistivity could be caused by the decrease of the stresses arising from preparation, and by the rearrangement of imperfections in the structure.

The subsequent rise in the annealing curve is caused by oxidation of the film. This can be seen in Figure 9(b). The island structure of the film remains unchanged but it is partially oxidized. An oxide phase can be seen on the diffraction pattern. There is a slight, pointed reflection on the diffraction pattern. This indicates the presence of a third phase which has greater crystals than the amorphous nickel and oxide phase. This phenomenon shows that, in addition to oxidation, another process is starting: separation of a Ni–P compound phase from the supersaturated solid solution of phosphorous dissolved in nickel. The separation of this compound phase causes a decrease in resistivity[9], and therefore, after reaching its maximum the resistivity starts to decrease. If the film is further annealed, the quantity of the amorphous nickel phase continuously decreases (Figure 9(c)), the quantity of the other two phases increases and, finally, the film consists only of nickel oxide and the third crystalline phase mentioned above (Figure 9(d)). Hence, during annealing, the amorphous nickel gradually turns to nickel oxide and a Ni–P compound phase. On the last part of the annealing curve, where a very small resistivity change can be measured, no amorphous nickel phase can be identified; there is in the film only a fine crystalline nickel oxide and a Ni–P compound phase in the form of greater crystals. This means that the resistivity decrease continues until the amorphous nickel base is transformed in the processes mentioned above.

In the last stage of annealing (Figure 9(d)), the diffraction pattern contains continuous, but slightly diffuse rings corresponding to a fine-crystalline nickel oxide phase. It is difficult to determine the structure of the crystalline phase from the pointed rings, which correspond to a very small quantity of large crystals because compounds with different composition (i.e. Ni_5P_2, Ni_7P_5, Ni_3P) have very similar lattice constants. If similarly prepared amorphous films are annealed in vacuum, Ni_3P and Ni_5P single crystals are formed[10]. Therefore it is possible that, when annealing in air, these two compound phases are present together with the nickel oxide. The change of the temperature coefficients of resistors as a function of annealing time is shown in Table 1. The

Table 1. Change of the TCR value as a function of the annealing time for annealing at 250°C and 300°C, respectively

Time of annealing in hours	Temperature of annealing 250°C TCR ppm/°C	300°C TCR ppm/°C
0	21	13
0.2	20	
0.3		29
1	4	
2.3		94
5	9	
10	−13	
17	−62	
19		1248

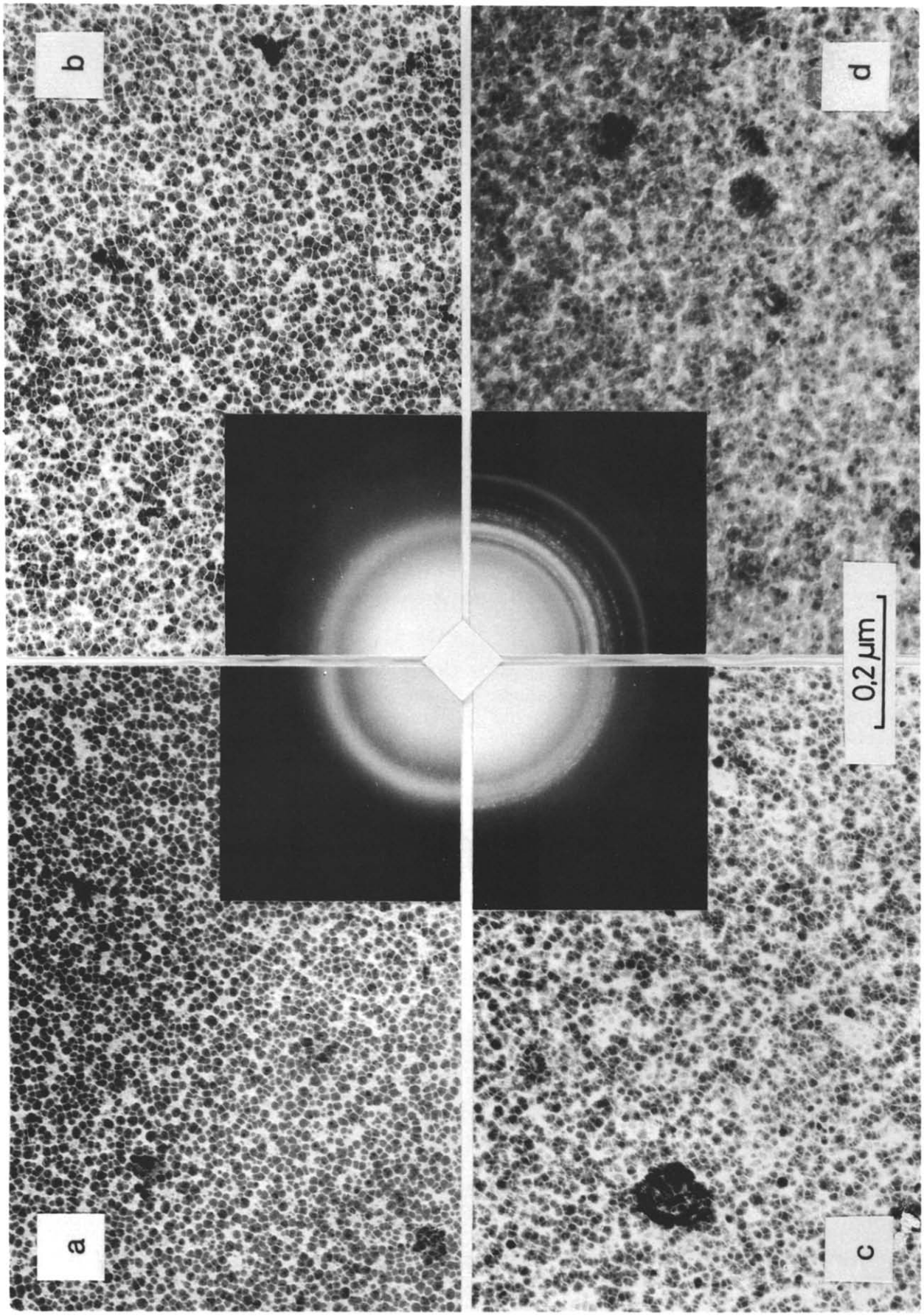


Figure 9. TEM pictures of nickel films after annealing at 300°C for different periods (a, b, c, d).

change of the TCR value confirms well the changes observed during structure investigations.

In the initial period of annealing the TCR value decreases (as can be seen by annealing at 250°C) and this decrease is followed by an increase. The decrease of the TCR value can be explained by an oxidation, and the increase of the TCR value after the maximum resistivity point of the annealing curve can be attributed to the formation of the crystalline phase mentioned above.

Conclusions

Properties of nickel resistor films prepared by electroless deposition have been investigated. It is suggested that the main parameters which determine the electrical properties of the films are: the pH of the bath, the deposition time and the bath temperature.

The structure of the film is, depending on the pH-value of the bath (i.e. the phosphorous content of the film), either amorphous or crystalline, and the structure influences the electrical properties.

The behaviour of the film during annealing at different temperatures has also been investigated. The structural changes during annealing were examined by transmission electron microscopy and electron diffraction. There are four different sections in the annealing curve, in which different processes are occurring. Initially the resistivity of the film is decreasing, as a result of structural rearrangement. Later the increase of resistivity results from the effect of oxidation. Together with the oxidation the formation of a Ni–P compound phase starts and, when this process prevails, the resistivity of the film starts to decrease. This process continues until the amorphous nickel phase is consumed by the processes mentioned above. Changes in the temperature coefficient of the resistivity of the film can be explained by these processes.

References

[1] W Goldie, Metallic Coating of Plastics, Vol I, p 39. Electrochemical Publications Ltd. Middlesex (1968).
[2] B G Bagley and D J Turnbull, *Appl Phys*, **39**, 5681 (1968).
[3] J P Randin, P A Maire, E Saurer and H E Hinterman, *J Electrochem Soc*, **114**, 442 (1967).
[4] M Schlesinger and J P Marton, *J Phys Chem Solids*, **20**, 188 (1968).
[5] E Vafaei-Makhsoos, E Thomas and L E Toth, *Metallurgical Transaction*, **9A**, 1449 (1978).
[6] E Vafaei-Makhsoos, *J Appl Phys*, **51**, 6366 (1980).
[7] A Cziráki, B Fogarassy, I Bakonyi, K Tompa, T Bagi and Z Hegedüs, MFKI-34 (1980).
[8] J P Marton and M Schlesinger, *J Electro Chem Soc*, **115**, 16 (1968).
[9] P J Cote and L V Meisel, *Phys Rev*, **B20**, 3030 (1979).
[10] A Barna, private communication.